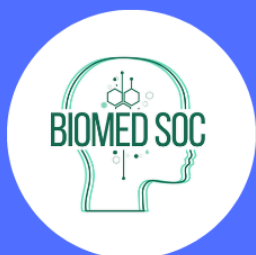


Bytes & Bolts Challenge



A collaboration between:



Overview

Welcome to **Bytes and Bolts**! Over the next 4 days you will have the opportunity to challenge yourself, collaborate with your team and showcase your skills in a high energy, innovative environment. At this multidisciplinary hackathon, you will be tasked with tackling complex problems and designing real-world solutions. By using your technical skills and creative thinking your team will be able to take home first place.

Team Structure

Teams may have 6 members, no more no less (unless no other options prevail).
For every different degree field within your team, beyond 1, your final score multiplier will increase by 3%.

Rough Schedule

Day	What	
Day 1 (22/06)	Opening Day	
Day 2 (23/07)	In-person Build day	Judging Periods
Day 3 (24/07)	In-person Build day (First Day of Marking)	1:00PM - 5:00PM
Day 4 (25/07)	In-person Build day (Last Day of Marking)	1:00PM - 4:00PM

Judging times may shift during the event; Listen out for announcements.

Marks are awarded at 100% of the value listed in the marking criteria if it is determined to have passed the requirements; otherwise, a mark of 0 will be awarded. Please be aware that there are no exceptions to this.

Challenges

This hackathon is made up of **3** challenges.

It is your choice as a team as to which challenges you want to attempt and how far you want to pursue each challenge. You can divide and conquer a bunch, or, full send just a few.

Awards

First Place

Highly Commended

Competitors that attend all 3 days will also receive a competition completion certificate.

Rooms

06.04.040 is our primary space, alongside the study rooms that border it. Teams are expected to work at their assigned desks when in these rooms.

Note: Areas in red are off limits/not to be used for the Hackathon



Shop, Print Queue & Leaderboard:

Our team this year has built a dedicated platform for buying components, uploading prints to the queue and submitting certain challenges for marking.

You will need your team's login (set with registration) to access the site. Do not share your password with other teams.

All teams begin the competition with the same, fixed amount of points.

Printing

Event printing services are only to be used for printing parts related to your projects.

Print requests will be handled in a first come first served manner.

Please do not submit unnecessary or duplicate requests.

Please also ensure that your prints will fit inside of the printers available, consult the print team for further details.

The print team may contact you before printing or reject your request at their discretion if they deem it unnecessary, unrelated to the competition or if it is impossible to be printed. .

Print sSettings:

By default all prints will be made at **10% gyroid infill**.

If you require your parts to be printed in specific orientations or with specific settings include details with files when you upload, and also, communicate directly with the printing team to ensure your requests are known and can be feasibly achieved.

Wearable Bionic Arm

Biomedical, Mechatronic & Accessible Design

Your friend Terevor the Tri-armed Trapezist lost their arm in a freak circus accident! As the nearest engineer you have been called in to construct a wearable prosthetic so the show can go on.

Teams must build a bionic arm to be worn by one of their members, with a focus on wearability and accessible design. While tasks require levels of biomimicry in prosthetic movement, the overall form is up to your team.

Criteria:

Points	Challenge	Outcomes	Level	Field/s
20	BASE CHALLENGE The arm must have the ability to grasp and lift objects. And which has a minimum of 2 single axis joints	Arm/Hand has at least 2 functioning single-axis joints. The Arm must be able to grab, lift and an object, for at least 5 seconds The Arm must be worn and stay attached to the user for the entire duration of the demonstration, and must not be helped or supported externally.	Base for completion	Mechanical / Mechatronic
5	Arm is able to interact with a touch display	User is able to interact with a smart phone display using their prosthetic and using the prosthetic to navigate the device can take a photo of a teammate. This must be filmed or demonstrated live to judges to earn points.	Moderate	Mechatronic
5	Arm is able to, (in a controlled manner) grip, lift and carry a 110g plastic water bottle from one marked position to the next.	The object must be at least 5cm off the table, carried 30cm from one marked point to the next (on the same table). Cannot be supported by any external force or help. The bottle must land within the marked circle and land upright.	Moderate	Mechanical/ Mechatronic
10	Arm is easily wearable and can be taken on & off within 5 minutes by the user, unassisted.	Arm can be removed and placed back on by the user without losing functionality or	Moderate	Biomedical, accessible design

		getting damaged. A table can be used for support. Up to 2 group members are able to spot the wearer to ensure the prosthetic isn't dropped or damaged but they may not assist the user in taking on or off the system. This must be filmed or demonstrated live to judges to earn points.		
5	Arm is able to pick up a sheet of paper from a flat surface	A flat A4 sheet of paper must be lifted from a smooth surface and remain held for at least 5 seconds. This must be filmed or demonstrated live to judges to earn points.	Low / Moderate	Mechanical
5	Arm is independently powered	Arm can perform the base challenge untethered to a wall or "external" power source.	Medium	Mechatronic
10	Prosthetic as able to move in 3 axis of motion.	Wrist, elbow and other joint/s, together creating motion in 3 directions. Multi-axial joints are permitted. The "elbow" must both flex and extend with a range of at least 45 degrees.	Hard	Mechanical/ Mechatronic
10	System is able to demonstrate a level of dexterity and coordination by moving between 3 marked points on the wall.	Arm must reach forward 5cm without the user physically stepping or leaning forward, and must reach 5 different points on a sheet of paper.	Moderate	Mechanical/ Mechatronic
10	The arm must grab and lift a specified "awkward" object to a marked spot on the table.	Outcome will be considered achieved if the object is lifted at least 5cm from the table, and is touching at least some part of the marked position when placed back down. The marked position will be 40cm away. Object must be picked up when the largest face is facing the table	Hard	Mechanical/ Biomedical

20	The prosthetic is able to be controlled entirely handsfree by the user.	User is able to demonstrate independent, hands-free control over the prosthetic system	Hard	Biomedical / Mechatronic
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Safety requirements

- No exposed wires, safely enclosed all moving parts.
- No internal voltages higher than 12V.
- Any tools (eg: Hot glue gun) must be used responsibly, and switched off when not in use

Supporting information:

Guides/tutorials

Learn solidworks - <https://my.solidworks.com/training>

- https://www.youtube.com/watch?v=pjeM4wBcD_0

- <https://www.youtube.com/watch?v=Ylquko4mXRM>

Introductions to slicing and 3d printing - <https://www.youtube.com/watch?v=GYfLrpJEr28>

- <https://www.youtube.com/watch?v=1i9nT1aDqgw>

Introductions to ardiono - <https://docs.arduino.cc/learn/starting-guide/getting-started-arduino/>

What are biosignals - <https://www.sciencedirect.com/topics/engineering/biosignal>

Home made EMG - <https://www.instructables.com/Muscle-EMG-Sensor-for-a-Microcontroller/>

Filtering Signals - <https://pmc.ncbi.nlm.nih.gov/articles/PMC10059683/>

- <https://lcamtuf.substack.com/p/the-101-of-analog-signal-filtering>

Required software

Your preferred IDE for programming.

We recommend arduino IDE if you don't have a preference already.

Some recommended parts / equipment

- An laptop
- Servos
- Microcontroller/s
- DC
- Cardboard / Foam core
- 3D printed PLA
- Multimeter
- Breadboard
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Time Attack

Software & Machine Learning

Speed is key as you race through two courses in a simulated environment. Reaching the top of the leaderboard demands perfection, with no room for human error. Not only will you be competing with your fellow peers but also with the judging panel, as you climb through the leaderboard to the top.

You and your team are tasked to achieve high speeds in time trials within the Trackmania Nations Forever game engine. Compete with other teams to reach the top of the leaderboard, and find innovative methods of racing through courses with the use of machine learning and AI. There will be two courses, one that is seen and distributed publicly to all teams, and one that is only available for limited runs during the testing period.

Criteria:

Points	Challenge	Level	Rough field
Points	Challenge	Lvl	Rough field
20	Complete the “seen” course available from challenge commencement, and publish a lap time to the leaderboard through the online platform.	Base completion level to score points	Software Engineering, Ai, Game Design
5	Complete the “seen” course available from challenge commencement without any human input, publishing a time to the leaderboard	Easy/Moderate	Software Engineering, Ai, Game Design
15	Complete the “seen” course available from challenge commencement without any human input, publishing a time to the leaderboard and beating the course time set by the judges.	Moderate	Software Engineering, Ai, Game Design
10~0	Top 10 teams on the leaderboard get points between 10 and 1, with faster times achieving higher points for the “seen” course.	Hard	Software Engineering, Ai, Game Design
20	Complete a full lap of a custom course unseen until testing, and publish a lap time to the leaderboard through the online platform. Testing will allow for 5 minutes of setup time and 3 attempts.	Moderate/hard	Software Engineering, Ai, Game Design
15	Complete a full lap of a custom course unseen until testing, publish a lap time to the leaderboard through the online platform and beat a course time set by the development team. Testing will allow for 5 minutes of setup time and 3	Hard	Software Engineering, Ai, Game Design

	attempts		
10~0	Top 10 teams on the leaderboard get points between 10 and 1, with faster times achieving higher points for the unseen course available on testing day	Hard	Software Engineering, Ai, Game Design
5	Set a real world speed record (top 10), within any tool assisted category for Trackmania Nations Forever.	Hard	Software Engineering, Ai, Game Design

Supporting information:

Guides/tutorials

Links to online examples and information for those who need it

History of ML in trackmania - <https://hallofdreams.org/posts/trackmania-1/>

History of shortcuts in trackmania <https://youtu.be/EVBfRuiabns?si=6dcscxl8l8CWB8qy>

Reinforcement learning https://www.youtube.com/watch?v=L_4BPjLBF4E

Tool assisted speed run information <https://donadigo.com/tmtas>

Game Installation Guide:

- Head to <https://www.trackmaniaforever.com/> containing the links to download the setup file
- Click TrackMania 1 demo
- Configure location of download

Map Installation Guide:

- Download map from teams folder (.gbx)
- Go to documents/TmForever/Tracks/Challenges/My Challenges
- Upload map into folder

Required software

TrackMania Nations Forever game (Available for **FREE**),

Some recommended parts / equipment

A computer (ideally running Windows or linux)

A New Sydney Tower Eye

Civil, Mechanical and Design

A catastrophic earthquake has struck Sydney, and the Sydney Tower Eye -with its cables over due for replacement- has been totally flattened!

It is your team's job to design and create a scale model that can replace the tower and all its previous qualities - height, innovation, and style. This design does not have to look identical to the existing tower, but it must meet the below minimum specifications.

Criteria:

Points	Challenge	Outcomes	Level	Field/s
20	Structure holds mass at required height and fits within maximum space (1.)	Structure supports 4 kilos at 500mm above the base Structure fits within 150x250mm and can be fastened to the shaketable Securing weight is teams responsibility Structure only uses materials within the allowed materials list	Base completion level to score points	Civil, Architecture
10	Structure uses 3D printed joints/connections for members, and all members/trusses are coffee stirrers (2.)	The majority of joints within the structure (i.e. where members or trusses connect) are 3D printed with PLA All members are coffee stirrers (meaning the rest of the structure is either 3D printed PLA, masking tape, hot glue, or string) NO members (or other load bearing elements) can be 3D printed A "joint" cannot be larger than 50x50x50mm	Moderate	Civil, Fabrication
10	Structure is prototyped and 3D modeled (3.)	Structure exists within a 3D modeling software that can accurately illustrate member construction, dimensions, and where forces act - this can include sketchup, Fusion etc	Easy	Civil, Architecture

		Structure is prototyped and tested on the shaketable - the prototype may contain entirely of Lego, for all other criteria the materials list must be abided by		
20	Structure has clear Aesthetic qualities (4.)	Structure has a well thought through design direction and has clear signs of effort and dedicated time Structure may contain materials not from the materials list for aesthetic purposes only (they must be non load bearing)	Moderate	Architecture
10	Structure has a mechanically geared and rotating observation platform (5.)	The platform can rotate and is geared with a mechanical advantage of at least 2:1 (gear reduction) - the platform CANNOT simply be a platform spun on a stick as it must include gears The rotating platform does not have to hold weight during testing, and can be removed for testing so long as the minimum criteria are still met	Easy	Mechanical
15	Rotating observation platform is mechanized (6.)	Observation platform rotates without the physical input of a person - it must be remotely activated	Moderate	Mechanical
15	Tower withstands Earthquake simulation (7.)	2 kilos are loaded onto the tower, and the tower is shaken using the earthquake table Structure should use anti earthquake dampening technologies (such as a mass dampener) It is the team's responsibility to secure the load and tower Structure is shook at 50% range for 20 seconds (14 swings), then 100% range for 8 seconds (8 swings) - intensity and frequency is subject to change	Hard	Civil

Safety requirements

- Ground beneath shaketable must be protected by impact resistant matting
- Students must keep distance from loading, especially keeping fingers and feet away from the shaketable
- Shaketable operators must use steel cap shoes
- Any tools (eg: Hot glue gun) must be used responsibly, and switched off when not in use

Supporting information:

Accepted Materials List:

- Wooden coffee stirrers
- 3D printed PLA (see above criterion 2.)
- Lego (only for prototyping, see above criterion 3.)
- Hot glue or super glue
- String (non-metal)
- Non-load bearing aesthetic materials/objects (see above criterion 4.)
- Non-load bearing componentry (see above criterion 5., 6.)
- Non-load bearing material for mass dampening (see above criterion 7.)

Guides/tutorials

Learn solidworks - <https://my.solidworks.com/training>

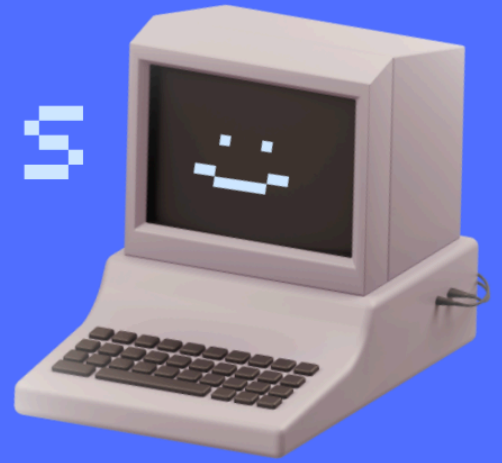
- https://www.youtube.com/watch?v=pieM4wBcD_0

- <https://www.youtube.com/watch?v=Ylguko4mXRM>

Introductions to slicing and 3d printing - <https://www.youtube.com/watch?v=GYfLrpJEr28>

- <https://www.youtube.com/watch?v=1i9nT1alDgw>

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